Nexora Retail | Operations Analytics Using SQL, Python & Matplotlib

An end-to-end business analytics portfolio project | **Reporting period:** Q1 2025 (January-March)

Dataset disclosure: Nexora Retail is a fictional business. This project uses a small, synthetic dataset created for learning and portfolio demonstration. The findings are descriptive, not statistically conclusive.

Project Overview

This project evaluates retail operations across warehouse fulfillment, delivery reliability, inventory replenishment, and product revenue. Operational records were extracted from a MySQL database, cleaned and analyzed with Python, and presented through a consolidated Matplotlib dashboard.

Objective: Identify operational performance gaps and translate data into practical business recommendations.

Business Questions

- Which warehouses achieve the highest and lowest on-time delivery rates?
- How does average dispatch-to-delivery transit time vary by warehouse?
- How does on-time delivery performance change across the months of Q1 2025?
- Which warehouse-product inventory records have reached their reorder levels?
- Which products account for the largest share of revenue?

Tools and Technologies

- **MySQL:** Relational storage and querying of operational tables.
- **SQLAlchemy + PyMySQL:** Database connection and SQL-to-Pandas data extraction.
- **Python / Pandas:** Cleaning, joining, filtering, grouping, aggregation, and monthly time-series analysis.
- **NumPy:** Conditional inventory classification with np.select().
- **Matplotlib:** Horizontal bar charts, stacked bars, a monthly line chart, Pareto analysis, and a combined dashboard using GridSpec.

Dataset and Scope

The synthetic dataset contains **30 orders**, of which **27 have valid completed-delivery records** for the delivery analysis, across **4 warehouses**. The product catalog contains **10 products**, and the inventory table contains **20 warehouse-product records**. The analysis focuses on **January-March 2025**.

Source tables:

- warehouses: warehouse identifiers, locations, operating costs, and capacity.
- products: product identifiers, categories, unit costs, and selling prices.
- orders: order dates, dispatch dates, delivery dates, promised dates, status, and shipping costs.
- order_items: order-product quantities and realized selling prices.
- inventory: stock quantity, reorder level, and last restock date by warehouse and product.

Metric definitions:

- **Completed orders:** Orders marked Delivered or Delayed with valid delivery and promised-delivery dates.
- **On-time delivery rate:** Completed orders delivered on or before their promised date / valid completed orders x 100.
- **Average transit time:** Mean number of days between dispatch date and delivery date.
- **Reorder required:** Stock quantity is greater than zero and less than or equal to reorder level.
- **Out of stock:** Stock quantity equals zero.
- **Product revenue:** Sum of quantity x unit selling price across order items.
- **Monthly reporting:** Completed orders are attributed to the month of their **order date**, not delivery date.

Dashboard

Dashboard image: dashboard/nexora_operations_dashboard.png (add image to GitHub repository)

Dashboard: delivery performance by warehouse, average transit time, inventory status, monthly on-time delivery, and product revenue Pareto analysis.

Key Findings

1. Overall delivery reliability requires attention

16 of 27 valid completed orders were delivered on time, resulting in an overall **59.26% on-time delivery rate**. This is a descriptive result for the sample and suggests that missed promised delivery dates warrant further investigation.

2. Delivery performance differs across warehouses

- **Warehouse 1:** 100.00% on time; average transit **2.00 days**; **7** completed orders.
- **Warehouse 2:** 50.00% on time; average transit **3.33 days**; **6** completed orders.
- **Warehouse 3:** 57.14% on time; average transit **3.43 days**; **7** completed orders.
- **Warehouse 4:** 28.57% on time; average transit **3.43 days**; **7** completed orders.

Warehouse 4 had the lowest observed on-time rate, while Warehouse 1 performed best. Warehouse 3 and Warehouse 4 had similar average transit times but different on-time rates, indicating that promised delivery windows and other process factors should also be examined. Each warehouse has only 6-7 completed orders, so comparisons should be treated cautiously.

3. Inventory replenishment needs are widespread

11 of 20 inventory records (55%) were classified as **Reorder Required**. Warehouse 1 had **2 of 5** records at or below reorder level; Warehouses 2, 3, and 4 each had **3 of 5**. No records were classified as **Out of Stock**.

A reorder flag is a replenishment signal, **not evidence of an actual stockout**. These counts do not by themselves establish demand, shortage severity, or lost sales.

4. Revenue is concentrated in a few products

The **top four products contribute approximately 80.58% of recorded product revenue**:

- **Laptop:** $225,000 (46.91%)
- **Monitor:** $72,000 (15.01%)
- **Office Desk:** $47,500 (9.90%)

- **Printer:** $42,000 (8.76%)

Revenue contribution is not the same as profit contribution. The Pareto chart highlights which products warrant closer monitoring for availability and sales concentration.

5. Monthly delivery performance is monitored

A monthly line chart tracks on-time delivery rates for **January, February, and March 2025**, using the order-creation month. The chart compares each month's rate against the overall **59.26%** benchmark. With only three months and a small number of orders, the visualization supports descriptive monitoring rather than long-term trend claims.

Business Recommendations

1. **Investigate Warehouse 4:** Review late orders, dispatch lead times, route constraints, carrier performance, and promised-delivery windows. Avoid assigning a root cause before checking order-level evidence.
2. **Prioritize replenishment reviews:** Examine the 11 flagged warehouse-product records. Combine reorder status with demand, supplier lead time, and safety-stock policies before deciding quantities.
3. **Protect high-revenue products:** Cross-check Laptop, Monitor, Office Desk, and Printer availability against their warehouse-level inventory positions. Prioritize based on both revenue exposure and actual demand.
4. **Track delivery reliability monthly:** Monitor on-time rate together with completed-order counts so percentage changes are interpreted in context.
5. **Expand the evidence base:** More orders and a longer observation period would make warehouse comparisons and operational recommendations more robust.

Analytical Workflow

```
MySQL operational tables
         |
         v
SQLAlchemy + PyMySQL connection
         |
         v
Pandas DataFrames
         |
         v
Data quality checks and datetime conversion
         |
         v
Delivery, inventory, product, and monthly analyses
         |
         v
NumPy classification + Pandas aggregations
         |
         v
Matplotlib charts + GridSpec dashboard
         |
         v
Business findings and recommendations
```

Repository Structure

```
Nexora-Retail-Operations-Analytics/
|
|-- README.md
|-- notebooks/
|   `-- nexora_operations_analysis.ipynb
|-- dashboard/
|   `-- nexora_operations_dashboard.png
|-- sql/
|   |-- database_schema.sql
|   `-- analysis_queries.sql
```

```
|-- data/
|   `-- [synthetic source tables as CSV files]
`-- requirements.txt
```

Only include files and folders that you have actually uploaded. The SQL analysis file and exported CSVs can be added as the project is finalized.

Reproducing the Analysis

1. Download or clone this repository.
2. Install the Python dependencies listed in requirements.txt.
3. Load the provided synthetic CSV files into Pandas, **or** recreate the MySQL tables using the provided SQL scripts and connect through SQLAlchemy.
4. Open notebooks/nexora_operations_analysis.ipynb and run the cells in order.
5. The final visualization section creates the dashboard image.

Connection safety: Do not commit database passwords, personal credentials, or a .env file containing secrets. A notebook intended for public review should include a CSV-based fallback so it can run without access to a local MySQL server.

Project Deliverables

- **Jupyter Notebook:** Data extraction, cleaning, calculations, visualizations, and interpretation.
- **Dashboard image:** Consolidated view of the five analytical visuals and summary KPIs.
- **README:** Business problem, approach, findings, limitations, and recommendations.
- **Optional SQL scripts:** Database schema, sample data, and equivalent SQL analysis queries.

Limitations

- The data is synthetic and does not represent a real company's operations.
- The delivery analysis uses only **27 valid completed orders**; warehouse comparisons are based on 6-7 orders each.
- The reporting period covers only **three months**.
- Reorder-level status does not measure actual stockouts or quantify lost revenue.
- Revenue is not profit; product profitability would require additional cost allocation.
- Observed differences and associations do not establish causal explanations.

Conclusion

The Nexora Retail project demonstrates an end-to-end workflow from relational data extraction to operational reporting. It identifies differences in warehouse delivery reliability, highlights replenishment flags, and shows that a small set of products accounts for most recorded revenue. The resulting dashboard translates technical analysis into business questions and practical next steps, while clearly acknowledging the dataset's limitations.

Skills demonstrated: MySQL | SQLAlchemy | PyMySQL | Pandas | NumPy | Matplotlib | Data Cleaning | KPI Development | Time-Series Aggregation | Business Analysis | Data Storytelling